

## BIOCHEMISTRY

## Enhanced native chemical ligation by peptide conjugation in trifluoroacetic acid

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Chemical ligation of peptides is increasingly used to generate proteins not readily accessible by recombinant approaches. However, a robust method to ligate “difficult” peptides remains to be developed. Here, we report an enhanced native chemical ligation strategy mediated by peptide conjugation in trifluoroacetic acid (TFA). The conjugation between a carboxyl-terminal peptide thiosalicylaldehyde thioester and a 1,3-dithiol-containing peptide in TFA proceeds rapidly to form a thioacetal-linked intermediate, which is readily converted into the desired native amide bond product through simple postligation treatment. The effectiveness and practicality of the method was demonstrated by the successful synthesis of several challenging proteins, including the SARS-CoV-2 transmembrane Envelope (E) protein and nanobodies. Because of the ability of TFA to dissolve virtually all peptides and prevent the formation of unreactive peptide structures, the method is expected to open new opportunities for synthesizing all families of proteins, particularly those with aggregable or colloidal peptide segments.

## INTRODUCTION

Chemical protein synthesis is a well-recognized strategy to produce proteins that would otherwise be difficult or impossible to obtain, such as posttranslationally modified and mirror-image proteins, for biochemical and biomedical research (1–7). The strategy commonly requires the assembly of peptide segments to form a full-length protein via techniques such as native chemical ligation (NCL) (8–10), ketoacid-hydroxylamine ligation (KAHA ligation) (11, 12), and serine/threonine ligation (STL) (13, 14), which have greatly advanced the field. In peptide ligation reactions, the choice of solvent is critical, as the solvent must effectively dissolve the peptide segments at the desired concentration (e.g., ~1 mM for NCL) and simultaneously expose their reactive moieties for ligation (5). NCL is typically conducted in neutral aqueous solutions containing chaotropic agents such as guanidinium chloride (Gn-HCl) or urea (1, 8), with organic solvents like dimethyl sulfoxide (DMSO)/H<sub>2</sub>O (12) and pyridine (Py)/acetic acid (13, 14) often used for KAHA ligation and STL, respectively. Besides, both KAHA and STL work under acidic conditions. These solvents can usually work well for the proteins composed of well-behaved hydrophilic peptides (15–19), but recent studies have highlighted the increasingly often encountered challenges associated with the ligation of “difficult” peptides, which are either too poorly soluble to reach the millimolar concentrations required or prone to form aggregates or other undesired secondary structures that may mask the ligation sites (20–28).

Trifluoroacetic acid (TFA), one of the most powerful and most commonly used solvents in peptide chemistry (29–32), can effectively dissolve virtually all peptides and prevent the formation of unwanted

secondary structures or aggregates (5, 6, 23, 33–35). We envisioned that TFA, if it can be used as the ligation solvent, could address the long-standing and fundamental challenge presented by the ligation of difficult proteins, thereby enabling more general and robust chemical synthesis of all families of proteins. In this context, we report a TFA-assisted peptide conjugation followed by intramolecular native chemical ligation (TAL) method, which entails the chemoselective condensation of any family of peptide segments (Fig. 1).

## RESULTS

## The enhanced native chemical ligation by peptide conjugation in TFA

In our recent study on the development of substrates for STL, we found that 1,3-propanedithiol can react rapidly with a peptide salicylaldehyde ester in TFA to form an S,S-propanedithioacetal (36). Inspired by this finding, we designed a peptide thiosalicylaldehyde thioester (**1a**) and a 1,3-dithiol-containing Cys-peptide (**2a**) and attempted their ligation in TFA. Peptide **1a** was prepared from the corresponding peptide hydrazide by oxidation with NaNO<sub>2</sub> (37), followed by thiolysis and acetal hydrolysis (42% isolated yield; fig. S1). Peptide **2a** bearing a removable 1,3-dithiol auxiliary group was also readily synthesized using our previously developed removable backbone modification (RBM) strategy (40% isolated yield; fig. S18) (38).

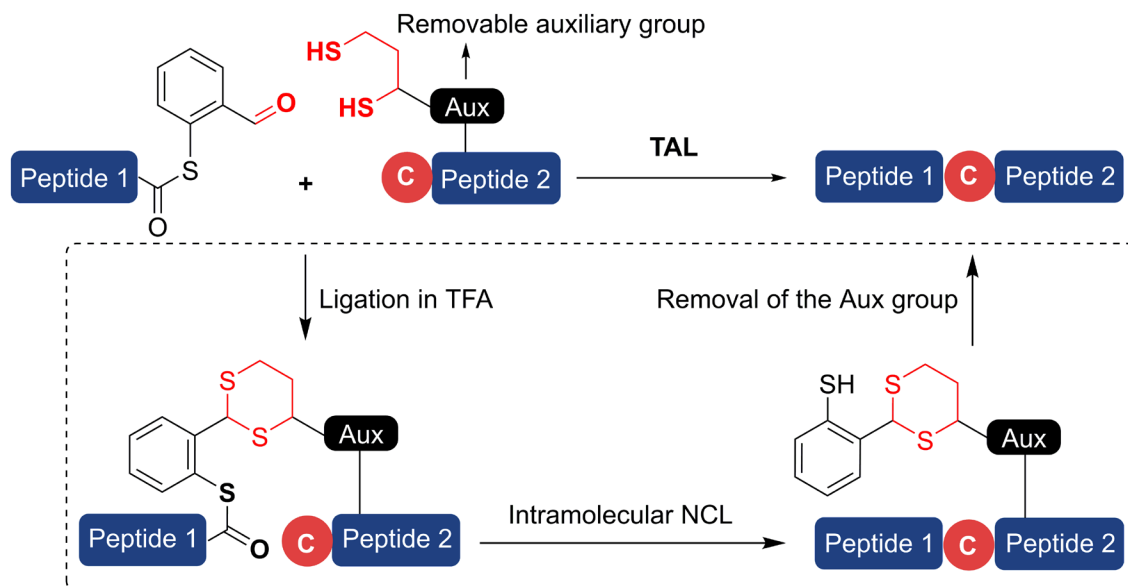
We dissolved **1a** (10 mM) and **2a** (10 mM) in TFA at room temperature. Reversed-phase high-performance liquid chromatography (RP-HPLC) analysis showed that **1a** and **2a** were completely and clearly consumed within only 30 s, accompanied by the formation of a new single peak **3a** (97% HPLC yield), the corresponding molecular mass (MM) of which (2469.66 Da) was consistent with that of the thioacetal intermediate (theoretical MM: 2470.03 Da; Fig. 2B). The TFA solution was then concentrated by N<sub>2</sub> gas blow-down and precipitated with cold diethyl ether to give solid **3a**, which was used without purification. **3a** (0.1 mM) was dissolved in a neutral phosphate buffer (H<sub>2</sub>O/CH<sub>3</sub>CN, v/v = 1:1; 20 mM phosphate, pH 7) to quantitatively afford **4a** (97% HPLC yield) within 1 min. The observed MM of **4a** (2384.17 Da) was consistent with

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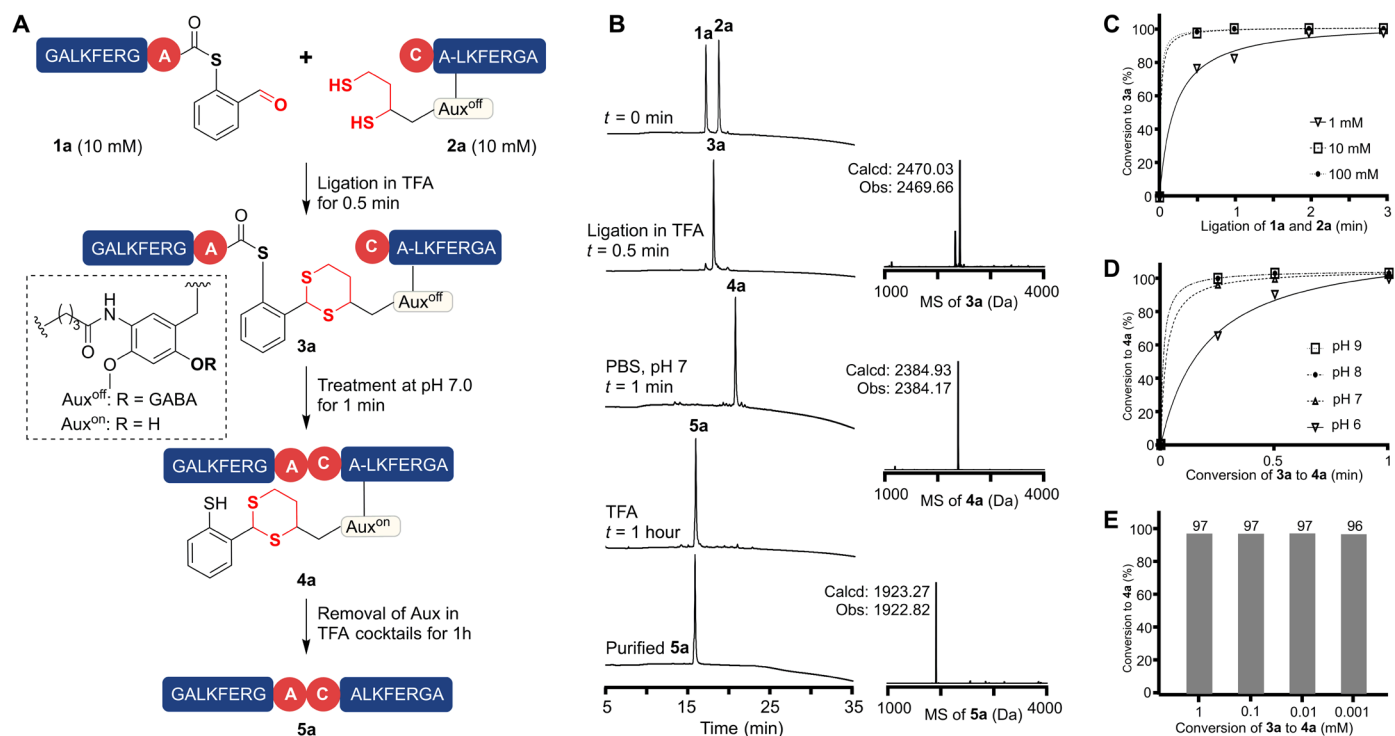
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**Fig. 1.** The principle of the TFA-assisted peptide conjugation followed by intramolecular native chemical ligation (TAL) method.



**Fig. 2.** The TAL method. (A) Synthesis of model peptide 5a by the TAL method. (B) HPLC and ESI-MS analyses. (C) Kinetics of the ligation of 1a and 2a in TFA at different reaction concentrations. (D) Kinetics of the conversion of 3a to 4a at various pH values. (E) Yields for the conversion of 3a to 4a at different concentrations.

the theoretical MM of the ligated peptide (2384.93 Da; Fig. 2B), and it was stable in the presence of 5% hydrazine for 30 min (fig. S24), suggesting that the thioester moiety in 3a had been converted to the amide bond of 4a. Last, 4a was dissolved in a TFA cocktail (TFA/PhOH/dithiothreitol (DTT)/H<sub>2</sub>O/thioanisole, 87.5/2.5/2.5/5/2.5, v/w/v/v/v) to remove the auxiliary (Aux) group and generate 5a within 2 hours (HPLC yield of 94%). The observed MM of 5a

(1922.82 Da) was consistent with that of the target product (theoretical MM: 1923.27 Da). The isolated yield of the final product 5a was 66% from 1a, further evidence of the feasibility and good efficiency of the TAL method.

In a control experiment, peptides 1a (1 mM), 2a (1 mM), and 2e (1 mM) were dissolved together in TFA. As expected, 1a only reacted with 2a, but not with 2e (fig. S25), demonstrating that the

aldehyde group of **1a** chemoselectively reacts with the 1,3-dithiol of **2a** in TFA, while the N-terminal Cys and the thiol group on the adjacent Cys remained inert. Comparison of the HPLC traces of **3a** and the ligation product of thioester **1q** with **2a** showed no detectable racemization at the ligation site (Table 1, entry 17, and figs. S26 and S27). The effect of the concentration of **1a** and **2a** in TFA on reaction time was also investigated (Fig. 2C and fig. S28): A reaction time of 30 s was sufficient to obtain HPLC yields of 97% and 98% at peptide concentrations of 100 mM and 10 mM, respectively, whereas 3 min and 30 min were required for HPLC yields of 97% at both 1 mM and 0.1 mM, respectively.

Table 1. The scope of the TAL method.

| Entry | 1X                 | 2Y                               | Product   | Overall Yield (%) |
|-------|--------------------|----------------------------------|-----------|-------------------|
| 1     | Ala( <b>1a</b> )   | Cys( <b>2a</b> )                 | <b>5a</b> | 94                |
| 2     | Arg( <b>1b</b> )   | Cys( <b>2a</b> )                 | <b>5b</b> | 91                |
| 3     | Glu( <b>1c</b> )   | Cys( <b>2a</b> )                 | <b>5c</b> | 90                |
| 4     | Gly( <b>1d</b> )   | Cys( <b>2a</b> )                 | <b>5d</b> | 94                |
| 5     | His( <b>1e</b> )   | Cys( <b>2a</b> )                 | <b>5e</b> | 90                |
| 6     | Ile( <b>1f</b> )   | Cys( <b>2a</b> )                 | <b>5f</b> | 95                |
| 7     | Leu( <b>1g</b> )   | Cys( <b>2a</b> )                 | <b>5g</b> | 92                |
| 8     | Lys( <b>1h</b> )   | Cys( <b>2a</b> )                 | <b>5h</b> | 91                |
| 9     | Met( <b>1i</b> )   | Cys( <b>2a</b> )                 | <b>5i</b> | 90                |
| 10    | Phe( <b>1j</b> )   | Cys( <b>2a</b> )                 | <b>5j</b> | 92                |
| 11    | Pro( <b>1k</b> )   | Cys( <b>2a</b> )                 | <b>5k</b> | 85                |
| 12    | Ser( <b>1l</b> )   | Cys( <b>2a</b> )                 | <b>5l</b> | 90                |
| 13    | Thr( <b>1m</b> )   | Cys( <b>2a</b> )                 | <b>5m</b> | 94                |
| 14    | Trp( <b>1n</b> )   | Cys( <b>2a</b> )                 | <b>5n</b> | 91                |
| 15    | Tyr( <b>1o</b> )   | Cys( <b>2a</b> )                 | <b>5o</b> | 93                |
| 16    | Val( <b>1p</b> )   | Cys( <b>2a</b> )                 | <b>5p</b> | 91                |
| 17    | D-Ala( <b>1q</b> ) | Cys( <b>2a</b> )                 | <b>5q</b> | 94                |
| 18    | Ala( <b>1a</b> )   | Cys( <b>2b</b> )                 | <b>5a</b> | 94                |
| 19    | Ala( <b>1a</b> )   | Cys( <b>2c</b> )                 | <b>5a</b> | 93                |
| 20    | Ala( <b>1a</b> )   | Gly <sup>Aux</sup> ( <b>2d</b> ) | <b>5r</b> | 93                |
| 21    | Val( <b>1p</b> )   | Gly <sup>Aux</sup> ( <b>2d</b> ) | <b>5s</b> | 90                |

Conversion of **3a** to **4a** can be carried out with a wide range of peptide concentrations from 0.001 to 1 mM and a broad pH range from 5 to 9, all with high HPLC yields (more than 96%; Fig. 2, D and E, and figs. S29 and S30). In addition, quantitative conversion of **3a** to **4a** (HPLC yields more than 95%) can be readily performed in various solvent conditions. This includes denaturing aqueous solutions containing agents such as guanidine hydrochloride (Gn-HCl) or urea, as well as various combinations of water (H<sub>2</sub>O) with organic cosolvents such as acetonitrile (CH<sub>3</sub>CN), Py/acetic acid mixture, dimethylformamide (DMF), DMSO, hexafluoroisopropanol, and 2,2,2-trifluoroethanol.

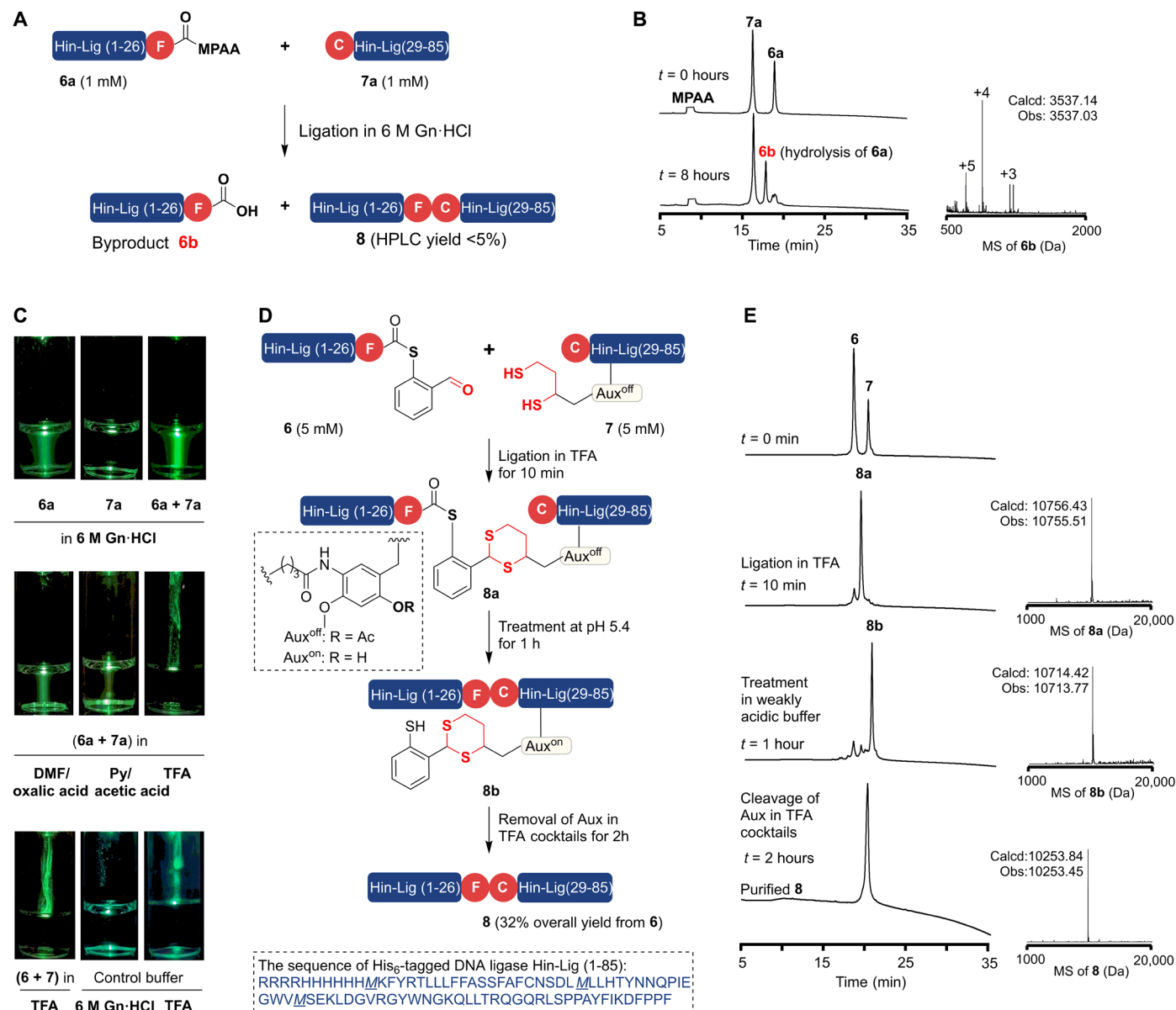
To investigate how the performance of TAL was affected by the location of the 1,3-dithiol modifications, the 1,3-dithiol group was introduced onto the amide between Lys<sup>4</sup>-Phe<sup>5</sup> or Gly<sup>8</sup>-Ala<sup>9</sup> sites in **2** to generate peptides **2b** and **2c**, respectively. **2b** and **2c** were separately ligated with **1a** in TFA to afford **3b** and **3c**, which were converted to the corresponding **4b** and **4c** by treatment with phosphate buffer (0.1 mM, H<sub>2</sub>O/CH<sub>3</sub>CN, v/v = 1/1; 20 mM phosphate, pH 7) for 1 min and then advanced to the desired product **5a** in HPLC yields of 94% and 93%, respectively, by exposure to TFA cocktails (TFA/PhOH/DTT/H<sub>2</sub>O/thioanisole, 87.5/2.5/2.5/5/2.5, v/w/w/v/v) for 2 hours (Table 1, entries 18 and 19, and figs. S32 and S33). These results show that the TAL method can tolerate different sites for the installation of the 1,3-dithiol group.

The scope of the TAL method was assessed via the ligation in TFA between **2a** and a series of model peptide thiosalicylaldehyde thioesters (**1b** to **1p**) bearing various C-terminal amino acids and having the general sequence GALKFERGX. The desired ligated peptides **5b** to **5p** except **5k** were obtained with >90% HPLC yields (Table 1, entries 2 to 16); **5k**, a peptide formed by the ligation of the highly sterically hindered Pro and Cys residues, formed in 85% HPLC yield (Table 1, entry 11). Moreover, **1a** and **1p** could ligate in TFA with **2d**, bearing an N-terminal Gly(Aux), to generate the desired target products **5r** and **5s** (HPLC yields of 93% and 90%, respectively; Table 1, entries 20 and 21, and figs. S49 and S50). These results show that the TAL method is highly tolerant of sterically hindered amino acids at the C-terminal and N-terminal sites.

The TAL method enabled the assembly of peptides prone to form colloidal particles

In previous work, we found that the peptide thioester Hin-Lig (1-27)-MPAA **6a** derived from *Haemophilus influenzae* DNA Ligase (Hin-Lig) failed to ligate with Hin-Lig (28-85) **7a** in phosphate buffer (pH 6.5) containing 6 M Gn-HCl, a denaturing solvent often used for NCL, because of the formation of colloidal particles (27). Although **6a** and **7a** did dissolve in the Gn-HCl-containing buffer, only hydrolytic by-product **6b**, but not ligation product **8** was detected (Fig. 3, A and B). An aqueous 6 M Gn-HCl solution of **6a** or a mixture of **6a** and **7a** illuminated with a 530-nm laser pointer exhibited an obvious Tyndall effect, indicating the formation of colloidal particles of **6a**, an effect also observed when **6a** was dissolved in DMF/oxalic acid and Py/acetic acid, solutions that are commonly used for KAHHA ligation and STL, respectively (Fig. 3C). In comparison, no Tyndall effect was detected for TFA solutions of **6a** and **7a** (Fig. 3C), indicating its ability to prevent the formation of undesired structures of **6a**.

Peptide thioester Hin-Lig (1-27)-TSAL thioester **6** and Hin-Lig [28-85, L<sup>35,Aux(LA)</sup>] **7** were prepared from the corresponding peptide hydrazide (46% isolated yield; fig. S53) and via a standard Fmoc-SPPS-based RBM strategy (12% isolated yield; fig. S55), respectively. Both peptides **6** (5 mM) and **7** (5 mM) easily dissolved in



**Fig. 3. Chemical synthesis of DNA ligase-Lig(1-85) by the TAL method.** (A) Native chemical ligation of **6a** and **7a** in 6 M Gn-HCl. (B) RP-HPLC and ESI-MS analyses of the ligation of **6a** and **7a**. (C) Tyndall effect of **6a**, **7a**, a mixture of **6a** and **7a**, and a mixture of **6** and **7** in various reaction solutions. (D) General synthetic route for DNA ligase-Lig(1-85) **8** by the TAL method. (E) RP-HPLC and ESI-MS analyses for the formation of **8**.

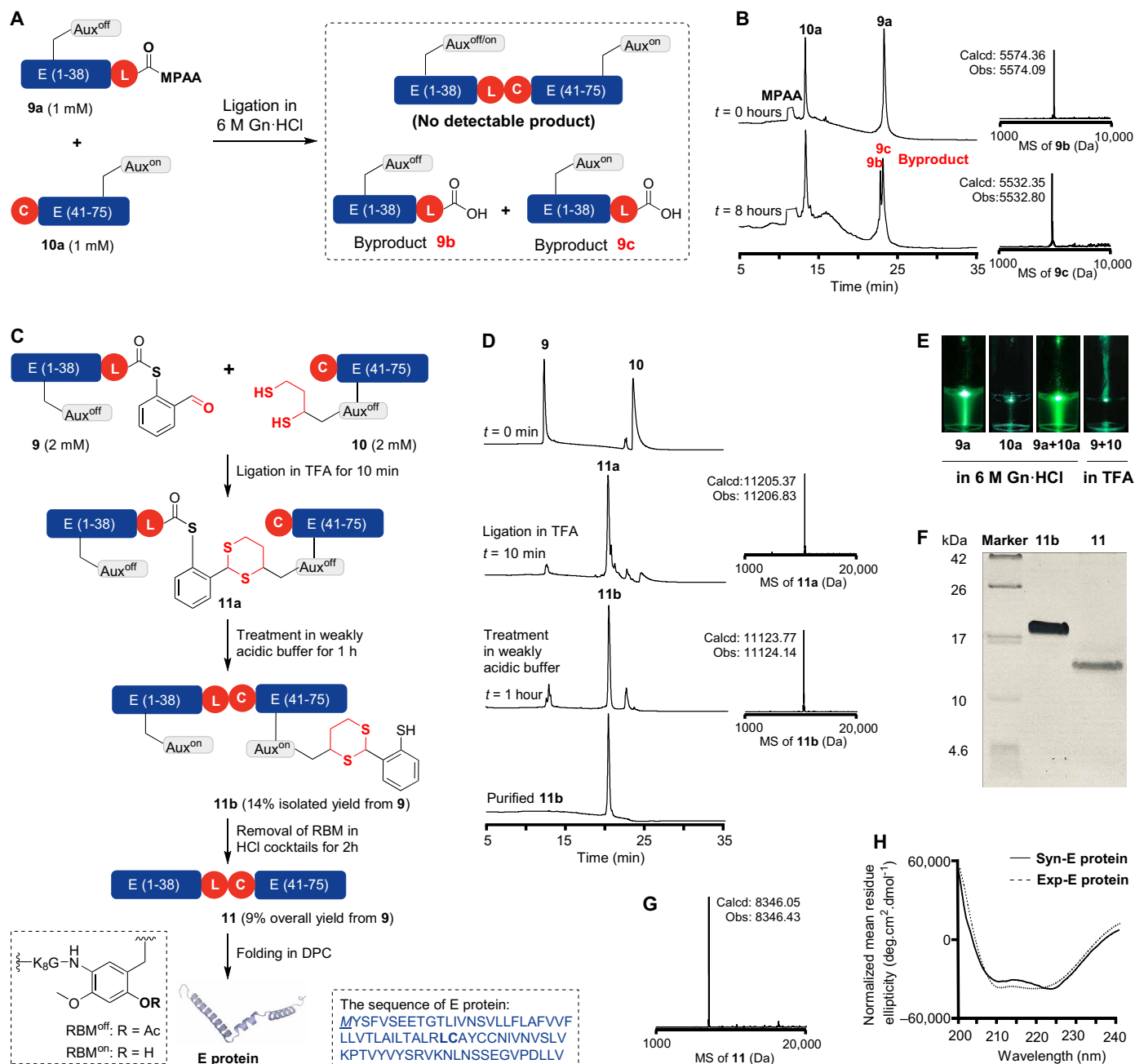
TFA to give a solution that did not exhibit a Tyndall effect. When peptides **6** (18 mg, 5 mM) and **7** (36 mg, 5 mM) were dissolved together in TFA, HPLC analysis showed that the reaction was complete within 10 min (Fig. 3, D and E) to give **8a**, electrospray ionization mass spectrometry (ESI-MS) analysis of which showed it to have a MM consistent with that of the peptide dithioacetal thioester (observed MM: 10,755.51 Da; theoretical MM: 10,756.43 Da; Fig. 3E). Posttreatment (comprising precipitation with chilled ethyl ether, treatment with weakly acidic buffer for 1 hour, and cleavage with TFA cocktails for 2 hours) of **8a** resulted in the production of the final product **8** (32% isolated yield starting from peptide **6**).

In summary, TFA shows better solubilizing and denaturing properties than solvents traditionally used for peptide ligation and allows

the efficient ligation of peptide segments refractory to canonical native chemical ligation.

### The TAL method promoted the chemical synthesis of membrane protein

To evaluate the potential of TAL to synthesize challenging proteins, we chose the single, 75-amino acid transmembrane protein severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) Envelope (E) protein as a target, which is implicated in viral budding, release, and the host inflammatory response (39). A synthesis of E protein would be useful because of its importance as a target for antiviral drugs and to support the study of how posttranslational modifications such as acetylation and glycosylation regulate its functions.



**Fig. 4. Chemical synthesis of E protein by the TAL method.** (A) Native chemical ligation between **9a** and **10a** in 6 M Gn-HCl. (B) RP-HPLC and ESI-MS analyses of the ligation of **9a** and **10a**. (C) The general synthetic route for E protein by the TAL method. (D) RP-HPLC and ESI-MS analyses for the formation of **11**. (E) The Tyndall effect of **9a**, **10a**, a mixture of **9a** and **10a**, and a mixture of **9** and **10** in different reaction solutions. (F) Tricine-SDS-PAGE of **11b** and **11**. (G) ESI-MS of **11**. (H) CD analyses of synthetic and recombinant E proteins in 100 mM DPC vesicles.

Initially, we applied the conventional NCL strategy to prepare the E protein by dividing it into two segments between Leu<sup>39</sup> and Cys<sup>40</sup>. Because of the hydrophobic nature of these segments, we temporarily installed a solubilizing Lys<sub>8</sub>-tag on the amide bond of Phe<sup>4</sup> and Ile<sup>46</sup> to give E (1-39, F<sup>4,Aux</sup>)-MPAA **9a** and E (40-75, I<sup>46,Aux</sup>) **10a** (figs. S60 and S62). Peptides **9a** and **10a** were successfully prepared via an Fmoc-SPPS-based RBM strategy in isolated yields of 44% and 33%, respectively. NCL of **9a** and **10a** was attempted by dissolving them in phosphate buffer (pH 7.0)

containing 6 M Gn-HCl at 1 mM. ESI-MS analysis showed that no ligation product was formed; instead, **9a** was completely hydrolyzed to form carboxylic acids **9b** and **9c** (Fig. 3, A and B). Further examination showed that a Gn-HCl solution of **9a** exhibited an obvious Tyndall effect, suggesting it to have formed colloidal particles (Fig. 4E). Therefore, although the RBM strategy can increase the solubility of hydrophobic peptides, it may not be sufficient to disrupt the formation of colloid structures, even in the presence of 6 M Gn-HCl.



We turned to the TAL method to address this challenge. Two peptides, E (1-39, F<sup>4,Aux</sup>)-TSAL thioester **9** and E [40-75, I<sup>46,Aux(LA)</sup>] **10** were synthesized with isolated yields of 18% and 16% (figs. S61 and S63). To our delight, both **9** (2 mM) and **10** (2 mM) were completely dissolved in TFA at a concentration of 2 mM without any detectable Tyndall effect (Fig. 4E) and underwent ligation within 10 min by RP-HPLC analysis to generate **11a** (Fig. 4, C and D), which was isolated as a solid pellet after concentration of the reaction mixture by N<sub>2</sub> blowdown and precipitation with cold ethyl ether. Treatment of **11a** with a weakly acidic (pH 5.6) buffer for 1 hour and hydrazine hydrate (NH<sub>2</sub>NH<sub>2</sub>/solvent = 1/100, v/v, pH 8 to 9) for 0.5 hours gave **11b**, which was dissolved in an HCl cocktail [1 M HCl/hexafluoroisopropanol and 5% triisopropylsilane (TIPS)] for 2 hours to generate the final product **11** with an overall yield of 9% starting from **9**. **11** was too hydrophobic for HPLC and was therefore analyzed by tricine-SDS-polyacrylamide gel electrophoresis (SDS-PAGE) instead; a clear band shift for **11** compared to its precursor **11b** (Fig. 4F) was observed. ESI-MS analysis of **11** showed that the observed MM (8346.43 Da) was consistent with the theoretical MM (8346.05 Da; Fig. 4G). Last, the folding of **11** was performed in dodecylphosphocholine (DPC) solution (100 mM DPC, 20 mM sodium phosphate, 50 mM NaCl, pH 5.5, 30°C). The circular dichroism (CD) spectrum showed two representative negative peaks at 208 nm and 222 nm (Fig. 4H) corresponding to structural features of  $\alpha$  helices, consistent with the previously reported CD spectrum of recombinant E protein (39).

### The TAL method allowed multiple-segment ligation to give nanobodies

Next, we used TAL to synthesize nanobodies (Nbs) composed of nine  $\beta$  strands. Nbs, which consist of the smallest naturally available antigen-binding VHH domains of antibodies but lack light chains, are a class of antibodies demonstrating improved tissue penetration and water solubility compared with traditional antibodies while retaining traditional antibody-like antigen-binding affinity and stability (40–42). However, recent studies have identified challenges in chemical synthesis of Nbs due to the inefficient ligation of  $\beta$  strand-rich peptides (43, 44).

We chose a 120-residue Nb that targets green fluorescent protein as a representative example (45). First, we tried the canonical ligation strategy. Nb was divided into three segments: Nb (1-52)-MPAA **12a**, Nb (53-94)-NHNH<sub>2</sub> **13a**, and Nb (95-120) **14a**, which were synthesized with isolation yields of 29%, 21%, and 41%, respectively (figs. S66, S68, and S70). We attempted to assemble these segments in an N-to-C direction. Peptides **12a** and **13a** were dissolved in the phosphate buffer (pH 7.0) containing 6 M Gn-HCl, without observable turbidity. However, HPLC and ESI-MS analyses revealed that only a small fraction (~5%) of the ligation product **15c** was produced, with most of the **12a** converted to its corresponding hydrolyzed by-product **12b** and intramolecular cyclization by-product **12c** (Fig. 5, A and B). We then discovered that 6 M Gn-HCl solutions of either **12a**, **13a**, or a mixture of the two all exhibited obvious Tyndall effects (Fig. 5C), indicating that both peptides had formed colloid structures.

We turned to use the TAL method to achieve the chemical synthesis of Nb. Three peptides Nb (1-52, C<sup>21,Acm</sup>)-TSAL thioester **12**, Nb [53-94, F<sup>67,Aux(LA)</sup>]-NHNH<sub>2</sub> **13**, and Nb [95-120, Q<sup>109,Aux(LA)</sup>] **14** were prepared. Peptide **12** was prepared with an isolated yield of 18% from the corresponding peptide hydrazide Nb (1-52, C<sup>21,Acm</sup>)-

NHNH<sub>2</sub>, in which the Cys<sup>21</sup> was temporarily protected by an Acm group (fig. S67). Peptides **13** and **14** bearing the 1,3-dithiol group at Phe<sup>67</sup> and Glu<sup>109</sup> were synthesized with isolated yields of 11% and 18%, respectively (figs. S69 and S71). Peptides **12** (2 mM), **13** (2 mM) and **14** (2 mM) were all fully soluble in TFA without evidence of any Tyndall effect, suggesting that TFA effectively inhibits the formation of peptide colloidal particles (Fig. 5C).

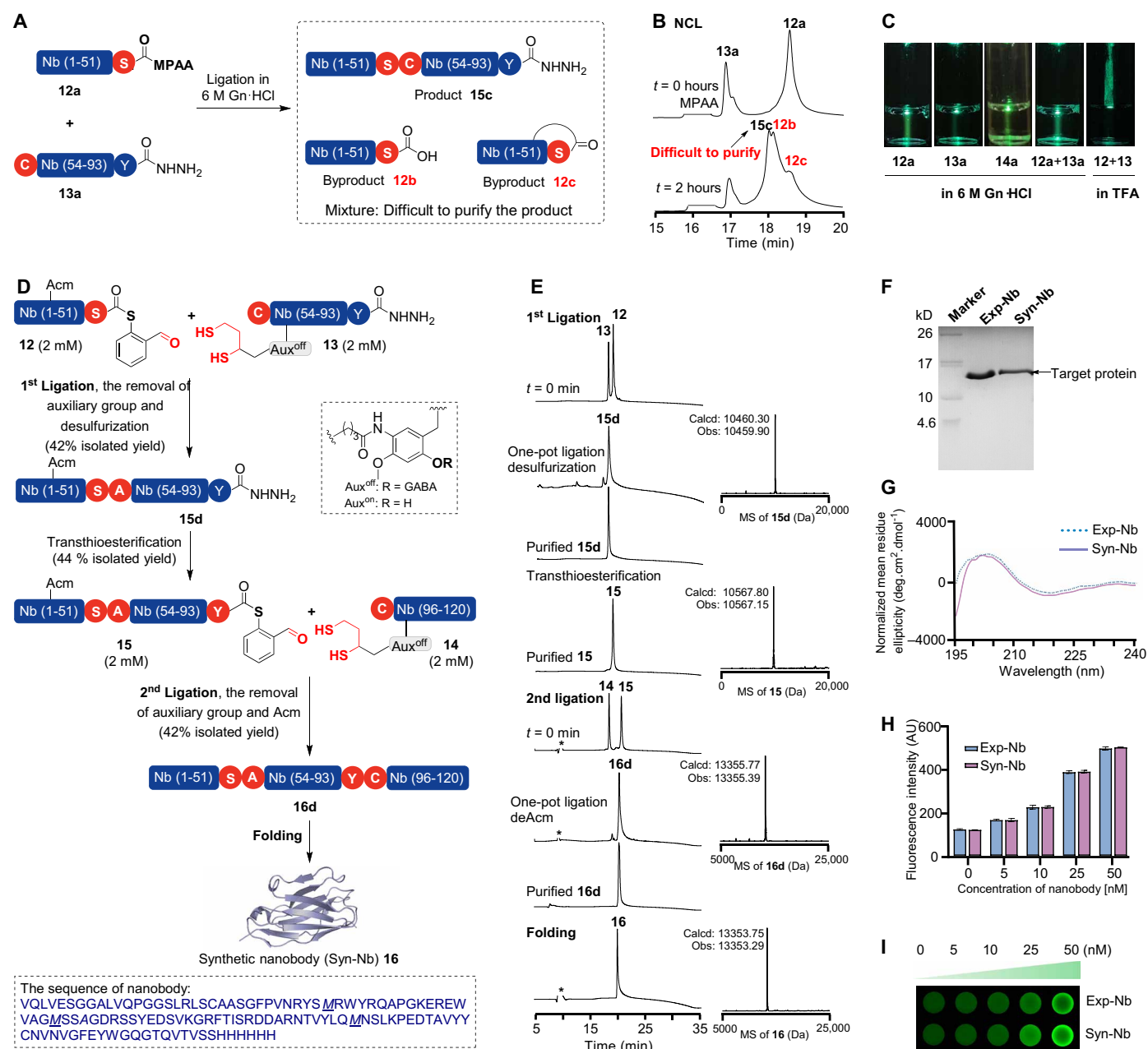
The assembly of full-length Nb was carried out in an N-to-C direction by two successive peptide ligations in TFA. First, **12** (2 mM) and **13** (2.1 mM) were dissolved in TFA, and the peptide dithioacetal thioester **15a** was formed within 5 min at 30°C, as confirmed by ESI-MS (observed: 10,459.90 Da; calculated: 10,460.30 Da). Solid **15a** was isolated after concentration and precipitation and then treated with weakly acidic H<sub>2</sub>O/CH<sub>3</sub>CN (v/v = 1/1) buffer (0.1 mM, Py/acetic acid, pH 5.6) to the backbone amide and TFA cocktails (TFA/TIPS/H<sub>2</sub>O, 95/2.5/2.5, v/v/v) to remove the auxiliary group (46). The resulting peptide underwent free-radical desulfurization by treatment with TECP/VA-044 to give the product **15d**, which was purified by HPLC (42% isolated yield starting from peptide **12**), and its molecular mass verified by ESI-MS (Fig. 5, D and E). Last, **15d** was converted to peptide thioester **15** in 44% isolated yield by the oxidation/thiolysis protocol.

Peptides **15** (1 mM) and **14** (1.2 mM) were ligated by mixing their respective TFA solutions. The ligation was complete within 5 min at 30°C to afford **16a**, the molecular mass of which was consistent with that of the expected peptide dithioacetal thioester (observed MM: 14,797.21 Da; theoretical MM: 14,797.53 Da). After concentration and precipitation, **16a** was treated with a weakly acidic H<sub>2</sub>O/CH<sub>3</sub>CN (v/v = 1/1) buffer (0.1 mM, Py/acetic acid, pH 5.6) and the TFA cocktails (TFA/PhOH/DTT/H<sub>2</sub>O/thioanisole, 87.5/2.5/2.5/5/2.5, v/w/w/v/v), followed by PdCl<sub>2</sub> (50 equiv.) to remove the Acm protecting group. Full-length Nb protein **16d** was purified by HPLC with an isolated yield of 44% starting from **15** and was verified by ESI-MS (Fig. 5, D and E).

Purified **16d** (0.2 mg/ml) was dissolved in an aqueous buffer [6 M Gn-HCl and 100 mM tris (pH 8.5)] at 4°C for 48 hours to form the intramolecular disulfide bond. Gn-HCl was removed by gradient dialysis. Subsequent purification by size exclusion chromatography yielded the final folded Nb **16** (isolation yield: 62%), which was characterized by ESI-MS and tricine-SDS-PAGE (Fig. 5F). CD analysis revealed that synthetic Nb **16** exhibited structural features including  $\beta$  sheets characteristic of recombinant Nb (Fig. 5G). We evaluated the binding of green fluorescent protein (GFP) and Nb by titrating purified wild-type GFP (WT GFP) with various concentrations (0, 5, 10, 25, or 50 nM) of either synthetic Nb **16** or recombinant Nb. Both synthetic and recombinant Nb induced fluorescence enhancement, and ca. fourfold enhancement of fluorescence intensity was observed when the concentration of Nb reached 50 nM, consistent with the previous report (Fig. 5, H and I) (43, 45).

## DISCUSSION

In summary, we have discovered that TFA can serve as an effective solvent for ligating virtually any peptide segments, including those that are poorly soluble in solvents like 6 M Gn-HCl aqueous buffer and/or prone to form colloid structures. In combination with native chemical ligation, this peptide conjugation strategy in TFA can conceptually address the long-standing and fundamental challenge presented by the ligation of increasingly often encountered difficult



**Fig. 5. Chemical synthesis of Nbs by the TAL method.** (A) Native chemical ligation between **12a** and **13a** in 6 M Gn-HCl. (B) RP-HPLC and ESI-MS analyses of the ligation of **12a** and **13a**. (C) Tyndall effect of **12a**, **13a**, **14a**, a mixture of **12a** and **13a**, and a mixture of **12** and **13** in different reaction solutions. (D) General route for the synthesis of an Nb by the TAL method. (E) RP-HPLC and ESI-MS analyses for the formation of **15** and **16** in TFA. (F) Tricine-SDS-PAGE of synthetic Nb and recombinant Nb. (G) CD analyses of synthetic GFP-Nb and recombinant Nb. (H and I) Fluorescence in vitro binding assay. Titration of synthetic Nb or recombinant Nb from 0 to 50 nM against 50 nM purified WT GFP. The fluorescence signal intensity of WT GFP was quantified using a multifunction microplate reader (H) and a laser scanner (I). Means and SD (error bars) of three independent experiments are shown.

proteins, thereby enabling more general and robust chemical synthesis of proteins. The TAL method is based on the selective and rapid reaction between a peptide thiosalicylaldehyde thioester and a 1,3-dithiol-containing peptide in TFA to give a thioacetal intermediate that can be easily converted to the final ligation product by a simple posttreatment involving concentration, precipitation, and treatment with buffer and TFA cocktails, with a similar as the prior thiol capture method and templated ligation (47, 48). The

scope of the TAL method was studied via the condensation of numerous model peptides, and its utility and practicality were exemplified by the successful preparation of several challenging proteins, including the E protein and an Nb. Collectively, these results establish the TAL method as a general, robust, and potentially invaluable method for the ligation of peptides and would expand the range of proteins now accessible by chemical synthesis. Studies to develop new types of TFA-compatible ligation reactions and optimize the

auxiliary groups used to install TFA-compatible reactive moieties onto peptides are ongoing in our laboratories and are expected to lead to a conceptually new category of peptide ligation technology; the results will be reported in due course.

## MATERIALS AND METHODS

### General materials and instruments

#### Reagents and materials

Rink amide resin and 2-Cl-Trt-Cl resin were bought from Nankai Hecheng Science & Technology Co., Ltd (Tianjin, China). Trityl-OH ChemMatrix resin and rink amide ChemMatrix resin were bought from PCAS BioMatrix Inc. The peptide synthesis tubes were bought from Synthware Glass Co. Ltd. All Fmoc amino acids and Boc amino acids were bought from GL Biochem (Shanghai, China). *O*-(7-azabenzotriazol-1-yl)-*N,N,N',N'*-tetramethyluronium hexafluorophosphate (HATU) and *O*-(6-chlorobenzotriazol-1-yl)-*N,N,N',N'*-tetramethyluronium 10 hexafluorophosphate (HCTU) were bought from GL Biochem (Shanghai, China). *N,N'*-diisopropyl-carbodiimide (DIC), ethyl cyanoglyoxylate-2-oxime (Oxyma), *N,N*-diisopropylethylamine (DIPEA), di-*tert*-butyl decarbonate, 4-dimethylaminopyridine (DMAP), *N*-Boc- $\gamma$ -aminobutyric acid (Boc-GABA-OH), lipoic acid (LA), sodiumborohydride ( $\text{NaBH}_4$ ), tetrahydroxydiboron [ $\text{B}_2(\text{OH})_4$ ], 4,4'-bipyridine, 4-mercaptophenylacetic acid (MPAA), tris(2-chloroethyl) phosphate (TCEP), trifluoroacetic acid (TFA), and triisopropylsilane were bought from Energy-Chemical (Shanghai, China). Pyridine, acetic acid (AcOH), 1,2-ethanedithiol (EDT), 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP), trifluoroethanol (TFE), and thioanisole were bought from J&K Scientific (Beijing, China). Guanidine hydrochloride (Gn-HCl), sodium phosphate monobasic ( $\text{Na}_2\text{HPO}_4$ ), *N,N*-dimethylformamide (DMF), dichloromethane (DCM), acetonitrile ( $\text{CH}_3\text{CN}$ ), methanol, anhydrous diethyl ether, hydrochloric acid (HCl), hydrazine hydrate aqueous solution ( $\text{NH}_2\text{NH}_2 \cdot \text{H}_2\text{O}$ ), di-*tert*-butyl decarbonate ( $\text{Boc}_2\text{O}$ ), acetic anhydride ( $\text{Ac}_2\text{O}$ ), and phenol were bought from Sinopharm Chemical Reagent.

#### High-performance liquid chromatography, mass spectrometry, and circular dichroism spectroscopy

Reverse-phase high-performance liquid chromatography (RP-HPLC) was performed on a Shimadzu Prominence HPLC System (Shimadzu Corp., Japan). Peptide analysis was performed on a YMC C4 (4.6 mm  $\times$  250 mm) column at a flow rate of 1.0 ml/min, and a YMC C4 (10 mm  $\times$  250 mm or 22 mm  $\times$  150 mm) column at a flow rate of 4.0 or 6.0 ml/min was used for peptide purification. Buffer A (0.1% TFA in water), buffer B (0.08% TFA in  $\text{CH}_3\text{CN}$  solution), and buffer C [0.08% TFA in the mixed solvents iPrOH- $\text{CH}_3\text{CN}$  (v/v, 1:1)] were used as the solvents. The solvent gradient was optimized for each peptide.

A Shimadzu Prominence HPLC System (Shimadzu Corp., Japan) with LCMS-2020 was used to record electrospray ionization mass spectrometry (ESI-MS) spectra. Applied Photophysics Pistar II-180 circular dichroism spectrometer was used to record circular dichroism spectra.

### General methods

#### Automated microwave peptide synthesis

All peptides were synthesized by using an automated microwave peptide synthesizer (CEM Liberty Blue). Rink amide AM resin was used to prepare C-terminal amide peptides, while 2-Cl-Trt-NHNH<sub>2</sub> resin was used to yield C-terminal hydrazide peptides. The resin was swelled in a mixture of DCM and DMF (5 ml + 5 ml) for 30 min. All other

amino acids were coupled using the standard microwave-assisted double coupling process. Each cycle involved a Fmoc deprotection step using 20% piperidine in DMF (1 min at 90°C) and an amino acid coupling step using a fourfold excess of 0.2 M Fmoc-protected amino acid, 0.5 M DIC, and 1.0 M Oxyma in DMF (twice for 10 min at 50°C for His and 2 min at 90°C for other residues). After peptide assembly, TFA cocktails (TFA/phenol/thioanisole/EDT/ $\text{H}_2\text{O}$ , 85/5/5/2.5/2.5, v/v/v/v/v) were used to cleavage peptides from the resin. The cleavage solution was collected and then concentrated by  $\text{N}_2$  blowing. The crude peptide was subsequently precipitated with cold diethyl ether and separated by centrifugation. Further analysis and purification were performed using RP-HPLC and ESI-MS.

#### Synthesis of auxiliary-containing peptides

The peptide assembly was performed using an automated microwave peptide synthesizer. After the removal of the Fmoc group of the amino acid that needs to be connected to the Aux, the remaining process requires manual handling, 2-hydroxy-4-methoxy-5-nitrobenzaldehyde [4 equivalent (equiv.)] in DMF was added to the resin and incubated for 60 min. Subsequently,  $\text{NaBH}_4$  (5 equiv.) in DMF was added to the resin for 5 min (twice). Thereafter, the resin was washed with the following solvents: DMF,  $\text{H}_2\text{O}$ , methanol, DCM, and DMF, respectively (five times).

The standard microwave-assisted double coupling protocol was used to assemble the following amino acids. Note that the amino group of the last amino acid should be protected with a Boc group for the subsequent Fmoc SPPS for Lys-tag or LA-tag.

After the peptide chain assembly,  $\text{B}_2(\text{OH})_4$  (20 equiv.) in DMF (4 ml for 0.1 mmol resin) and 4,4'-bipyridine (0.25 equiv.) in DMF (1 ml for 0.1 mmol of resin) were added to the resin and reacted for 20 min (twice). This step transformed the nitro into the free amino group.

For the introduction of a LA-tag, LA coupling was carried out by adding a solution of the LA (10.0 equiv.), HATU (9.8 equiv.), and DIEA (20 equiv.) in DMF to the resin for 1 hour (twice) at 30°C. In the case of Lys<sub>8</sub>-tag, a solution of Fmoc-Gly-OH (8.0 equiv.), HATU (7.6 equiv.), DMAP (0.8 equiv.), and DIEA (16.0 equiv.) in DMF were added to the resin for 1 hour (two times) at 30°C for the coupling of Fmoc-Gly-OH, followed by standard double coupling using the microwave method to introduce seven Fmoc-Lys(Boc)-OH and one Boc-Lys(Boc)-OH, respectively. For the LA-Lys<sub>6/8</sub>-tag, a solution of Fmoc-Gly-OH (8.0 equiv.), HATU (7.6 equiv.), DMAP (0.8 equiv.), and DIEA (16.0 equiv.) in DMF were added to the resin for 1 hour (two times) at 30°C for the coupling of Fmoc-Gly-OH, followed by a standard double coupling using the microwave method to incorporate either six or eight Fmoc-Lys(Boc)-OH residues. Then, LA was carried out by adding a solution of the LA (10.0 equiv.), HATU (9.8 equiv.), and DIEA (20.0 equiv.) in DMF to the resin for 1 hour (twice) at 30°C. For all tags, the resin was treated with 10 ml of 20% (v/v) of piperidine solution (DMF) reacting for 60 min at room temperature to remove all the amino acids attached to the Aux and subsequently release the 2-OH group. Notably, because of the potential of piperidine to cleave the disulfide bond in LA, we need to add DIEA (0.5 ml) in DMF to the resin for 30 min to promote the reformation of the disulfide bond for LA-tag and LA-Lys<sub>6/8</sub>-tag.

The 2-OH group of the auxiliary was then capped using  $\text{Ac}_2\text{O}$  or Boc-GABA-OH as follows, which enabled the auxiliary groups to withstand TFA cleavage: (i) a solution of 10 ml of  $\text{Ac}_2\text{O}$ /DIEA/DMF (1:1:8, v/v/v) to the resin for 30 min at 30°C; and (ii) Boc-GABA-OH (10 equiv.), DIC (10 equiv.), Oxyma (10 equiv.), and DMAP (1 equiv.) were dissolved in DMF and added to peptide resin for 20 min (two times) at 90°C. Last, a standard TFA cleavage step was



carried out to remove the side chain protecting groups and release the auxiliary-modified peptides.

### Preparation of peptide-MPAA

Peptide hydrazide (1 equiv.) was dissolved in 0.2 M phosphate buffer containing 6 M Gn-HCl (pH 3.0) and was oxidized by adding NaNO<sub>2</sub> (10 equiv.) to the solution for 20 min at −15°C to generate the corresponding peptide azide. To convert peptide azide into its thioester, MPAA (30 equiv.) was then added to the solution for 10 min, and the pH was adjusted to 5–6. Upon completion of the thioester transfer, TCEP (50 equiv.; pH 5) was added to the solution for 10 min to reduce the reaction. Then, the pH of the solution was adjusted to 1, and the MPAA was extracted with chilled ether (three times) before the peptide was separated by RP-HPLC.

### Preparation of peptide-TSAL thioester

Peptide hydrazide (1 equiv.) was dissolved in 0.2 M phosphate buffer containing 6 M Gn-HCl (pH 3.0) and was oxidized by adding NaNO<sub>2</sub> (10 equiv.) to the solution for 20 min at −15°C to generate peptide azide. To convert peptide azide into its thioester, 2-(1,3-dioxolan-2-yl) benzenethiol (30 equiv.) in CH<sub>3</sub>CN was then added to the solution for 10 min, and the pH was adjusted to 5–6. Notably, the volume ratio of aqueous buffer to CH<sub>3</sub>CN in the final reaction solution was 2 to 1. Upon completion of the thioester transfer, TCEP (50 equiv.; pH 5) was added to the solution for 10 min to reduce the reaction. Then, the pH of the solution was adjusted to 1, and the 2-(1,3-dioxolan-2-yl) benzenethiol was extracted with ice ether (three times) before the peptide was separated by RP-HPLC.

### Native chemical ligation

Peptide hydrazide (1 equiv.) was dissolved in 0.2 M phosphate buffer containing 6 M Gn-HCl (pH 3.0) and was oxidized by adding NaNO<sub>2</sub> (10 equiv.) to the solution for 20 min at −15°C to generate peptide azide. After that, the phosphate solution of MPAA (50 equiv.) and N-terminal Cys peptide (1 to 1.2 equiv.) was added for subsequent native chemical ligation (pH 6.5 to 7.0; 30°C). The ligation was tracked with analysis of RP-HPLC and ESI-MS. Once the ligation was accomplished, the ligation system was reduced by the addition of 0.1 M neutral TCEP 30 solution (100 equiv.) for 10 min.

### The TAL method

The ligation of peptide-TSAL thioester (1 to 10 mM) and the 1,3-dithiol-containing peptide (1 to 10 mM) was carried out in TFA solution at room temperature. The ligation was tracked with analysis of RP-HPLC and ESI-MS. Upon completion of the ligation, the TFA solution was concentrated by blowing N<sub>2</sub> and precipitation with cold diethyl ether and then afforded peptide-TSAL-dithioacetal thioester. This intermediate is subsequently characterized by RP-HPLC and ESI-MS. Then, postligation treatment of peptide-TSAL-dithioacetal thioester (0.1 to 1 mM) was performed under phosphate buffer (H<sub>2</sub>O/CH<sub>3</sub>CN, v/v, 1/1; 20 mM phosphate; pH 5.5 to 7) solution and took place rapidly and quantitatively to give native amide peptide. Last, the solution can be directly freeze-dried for the preparation of the native backbone peptide by treatment with TFA cocktails to remove the Aux group within 1 to 4 hours at 30°C. The final product was purified by RP-HPLC, and its identity was verified by ESI-MS.

### The Tyndall effect

The peptide was dissolved in ligation buffer (6 M Gn-HCl or TFA) in a glass vial. The solution was illuminated from the bottom of bottle using a laser pointer (~530 nm) to observe whether there was a bright laser beam. The observation of a bright laser beam led to the conclusion that the peptide had formed colloidal particles.

### Free radical desulfurization

The peptide (1.0 μmol) was dissolved in the desulfurization buffer [700 μl; 6.0 M Gn-HCl, 0.2 M Na<sub>2</sub>HPO<sub>4</sub>, and 0.5 M TCEP (pH 6.9)]. Subsequently, 70 μl of tBuSH and 700 μl of VA-044 solution (0.1 M in water) were added, and the reaction mixture was stirred at 37°C for 3 to 24 hours. The reaction was monitored using RP-HPLC and ESI-MS.

### Removal of the AcM group

The peptide containing the AcM group (1 mM) was dissolved in the aqueous buffer [6.0 M Gn-HCl and 0.2 M Na<sub>2</sub>HPO<sub>4</sub> (pH 7.2)] and treated with PdCl<sub>2</sub> (50 equiv.) for 15 min at 30°C to remove the AcM group. Upon completion of the reaction, 200 equiv. of dithiothreitol (DTT) was added to quench the reaction and precipitate free palladium from the reaction mixture. After centrifugation, the supernatant solution was collected and purified using semipreparative HPLC.

### Removal of the auxiliary group

The auxiliary-containing peptide was dissolved in TFA cocktails (TFA/PhOH/DTT/H<sub>2</sub>O/thioanisole, 87.5/2.5/2.5/5/2.5, v/w/w/v/v) and incubated at 37°C for 1 to 5 hours to remove the auxiliary groups. Note that the auxiliary groups can also be cleaved using HFIP and 5% triisopropylsilane with either 0.1 or 1 M HCl. After the cleavage, the resulting solution was concentrated using N<sub>2</sub> blowing and precipitated with chilled diethyl ether to yield the target peptide that was separated by centrifugation and then purified by RP-HPLC. The insoluble crude solid powder that could not be purified by HPLC underwent an additional washing step with water three times to eliminate the watersoluble Lys-tag. The final product was analyzed by ESI-MS and/or tricine SDS–polyacrylamide gel electrophoresis.

### Folding of SARS-CoV-2 envelope

The chemically synthetic SARS-CoV-2 envelope (E) (0.2 mg/ml) was folded under 20 mM sodium phosphate (pH 5.5) (100 mM diphenylamine carboxylate and 50 mM NaCl) at 30°C for 24 hours.

### Folding of Nb

The full-length Nb (10 mg) was dissolved in 50 ml of aqueous buffer [6.0 M Gn-HCl and 100 mM tris (pH 8.5)] and incubated at 4°C for 2 days to form an intramolecular disulfide bond. The folding process was then carried out by stepwise dialysis to obtain folded Nb, which was subsequently purified by gel filtration chromatography using a Superdex 200 10/300 GL column (GE Healthcare) with tris buffer (100 mM; pH 8.5).

### Fluorescence binding assay

The fluorescence binding assay was carried out by titrating 50 nM purified wild-type green fluorescent protein (WT GFP) with 0 to 50 nM of the chemically synthesized GFP Nb or the recombinant GFP Nb. The emission intensity of WT GFP was quantified using a laser scanner (Typhoon 9410, GE Healthcare; excitation, 488 nm) and a multifunction microplate reader (SpectraMax iD5, Molecular Devices; excitation, 470 nm; excitation, 510 nm).

### Cloning and purification of E protein, WT GFP, and Nb

The plasmid for expression was obtained by inserting the DNA sequences of E protein, WT GFP, or Nb into the pET-28a(+) plasmid using the restriction endonucleases Nco I and Xho I. GenScript Biotech (Nanjing, China) synthesized the genes for E protein, WT GFP, and Nb, which were then expressed and purified in the same manner. The BL21(DE3) *Escherichia coli* cells were transformed with the plasmid. The *E. coli* cells containing the expression vector were inoculated (1:100 dilution) into LB medium with kanamycin (100 μg/ml). Protein expression was induced with 1.0 mM isopropyl-β-D-thiogalactopyranoside at 37°C until the optical density at 600 nm

reached 0.6 to 0.8. After isopropyl- $\beta$ -D-thiogalactopyranoside treatment, the *E. coli* cells were able to express proteins at 18°C for 16 to 24 hours. The cells were then harvested and lysed by ultrasonication (30% power, 5-s on, 5-s off, 60 min) in 0.1 M Tris buffer [0.5 M NaCl and 1 mM phenylmethylsulfonyl fluoride (pH 8.0)]. The supernatant obtained after centrifugation was purified using a Ni-nitrilotriacetic acid (NTA) column. The elution from the Ni-NTA purification was combined and further purified using size exclusion chromatography on a Superdex 200 10/300 column (GE Healthcare). Last, the purified WT GFP or Nb was concentrated, flash-frozen in liquid nitrogen, and stored at –80°C for future use. For E protein, the elution from the Ni-NTA purification was subjected to cleavage of the His-SUMO tag by treatment with Ulp1 protease and dialyzed in dialysis buffer [50 mM Tris-HCl, 100 mM NaCl, and 2 mM DTT (pH 8.0)] at 4°C for 16 hours. The protein solution was filtered through Ni-NTA resin to collect the effluent, which was then purified by size exclusion chromatography to give the final product. Expressed sequences are as follows: E protein (His-SUMO-E protein), MGSSHHHHHSGSLVPRG-SASMSDSEVNQEAKPEVKPEVKPETHINLKVSDGSSEIFFKIK-KTTPRLRLMEAFARQKEMDSLRFLYDGIRIQADQTPEDLD-MEDNDIIEAHREQIGGMYSFVSEEIGTLIVNSVLLFLAFVVFLL-VTLAILTALRLCAYCCNIVNVLVKPSFYVYSRVKNLSSRVPDLLV; WTGFP, MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTCLKFICTTGKLPVPWPTLVTTFSYGVQCFSRYPDHMKQHDFEKSAMPEGYVQERTIFFKDDGNYKTRAEVKFEGDITLVNRIELKGIDFKEDGNILGHKLEYNNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDMVLLFVTAAGITHGMDELYKHHHHHH; Nb, MVQLVESGGALVQPGGSLRLSCAASGFPVNRYSMRWYRQAPGKEREWVAGMSSAGDRSSYEDSVKGRFTISRDDARNTVYLQMNSLKPEDTAVYYCNVNVGFEYWGQGTQVTVSSHHHHHH.

## Supplementary Materials

### This PDF file includes:

Supplemental Text

Figs. S1 to S75

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